



Introduction to SQL Commands and Their Categories

Lecture 3 · MySQL (SQL/PL-SQL) · BCA Program

PRESENTED BY RENUKA PARMAR

Previous Lecture Recap

Database Concepts

What is a database, DBMS vs RDBMS, and how data is organized

MySQL Overview

Introduction to MySQL as a relational database management system

Where We're Headed

Today we dive into how SQL commands are structured and categorized



Learning Objectives

By the end of this lecture, you will be able to:

01

Identify the five SQL command categories

DDL, DML, DQL, DCL, and TCL — and their core purpose

03

Map commands to real-world tasks

Apply each category to a practical Student Management System scenario

02

Distinguish between categories

Understand what makes each category unique and when to use it

04

Avoid common beginner mistakes

Recognize and correct the most frequent SQL categorization errors



Why Categorize SQL Commands?

SQL has dozens of commands. Without categories, it becomes overwhelming. Grouping them by **purpose and behavior** makes learning structured and logical.

→ Clarity

Know exactly what a command does before you write it

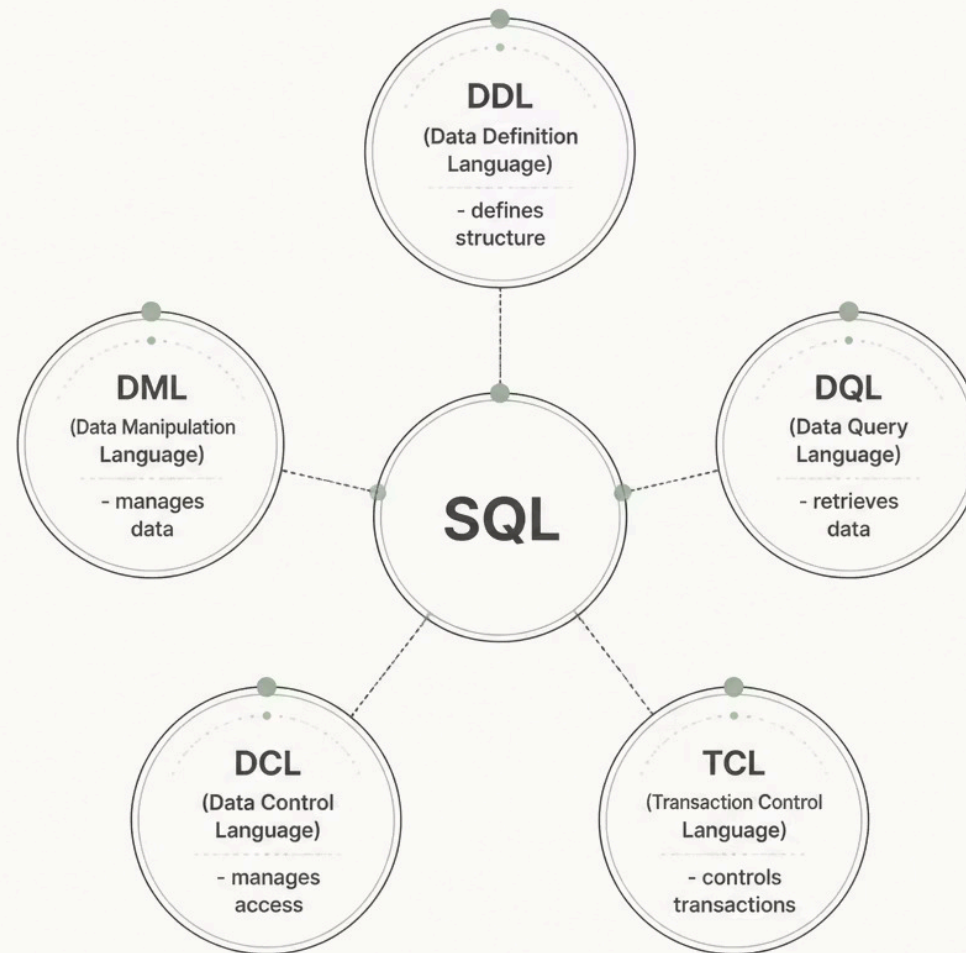
→ Control

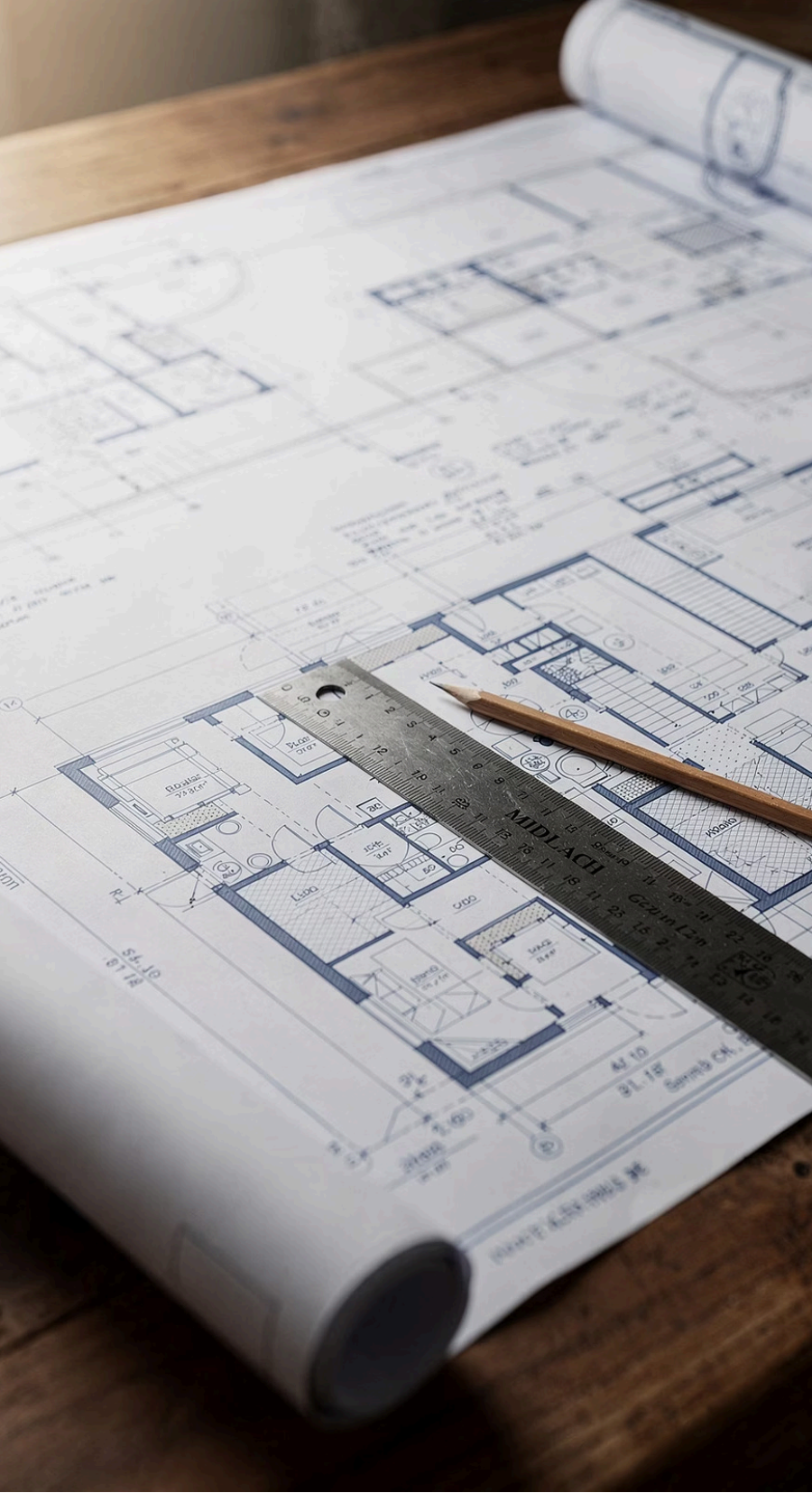
Different categories require different permissions and privileges

→ Safety

Some categories are reversible; others are permanent — crucial to know the difference

Overview of SQL Command Categories





DDL — Data Definition Language

Purpose

DDL commands **define and manage the structure** of database objects like tables, indexes, and schemas. Changes made by DDL are **auto-committed** — they cannot be rolled back.

- CREATE — builds a new table or object
- ALTER — modifies an existing structure
- DROP — permanently deletes an object
- TRUNCATE — removes all rows, keeps structure
- RENAME — changes the name of an object

Real-World Example

Building a **Student Management System**: You first *create* the Students table, then *alter* it to add a new column like "Email", and eventually *drop* a table that's no longer needed.



DDL shapes the skeleton of your database.

DML — Data Manipulation Language

Purpose

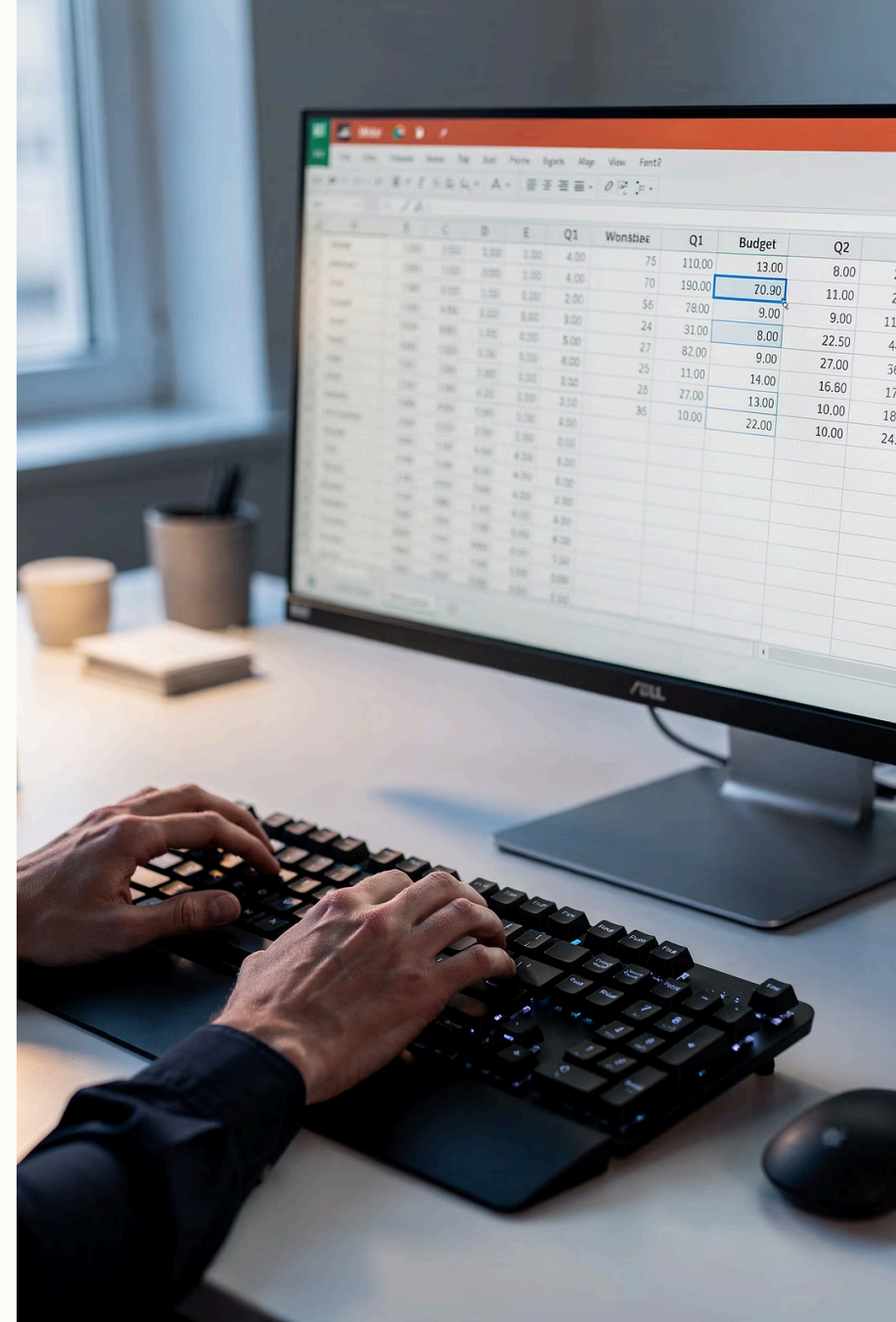
DML commands **insert**, **update**, and **delete data** within existing tables. Unlike DDL, DML changes can be **rolled back** using TCL commands before committing.

- INSERT — adds new records to a table
- UPDATE — modifies existing records
- DELETE — removes specific records

Real-World Example

In a Student Management System: *Insert* a new student's enrollment details, *update* their address when they move, or *delete* their record if they withdraw from the course.

- ❏ DML manages the day-to-day content of your database.



DQL — Data Query Language

Purpose

DQL is used to **retrieve and read data** from the database. It does not modify any data — it simply fetches records based on conditions you define.

- SELECT — the single most powerful SQL command

SELECT supports filtering (WHERE), sorting (ORDER BY), grouping (GROUP BY), and joining multiple tables.

Real-World Example

In a Student Management System: Use SELECT to *fetch all students who scored above 75%*, or retrieve a list of students enrolled in a specific course during a particular semester.

- ❏ DQL is read-only — the safest category for end-user access.





DCL — Data Control Language

Purpose

DCL commands control **who can access the database** and what actions they are permitted to perform. Managed by Database Administrators (DBAs).

- GRANT — gives a user specific access privileges
- REVOKE — removes previously granted privileges

Real-World Example

In a Student Management System: The principal is *granted* full access to all records. A student is only granted SELECT access to view their own profile. If a student graduates, their access is *revoked*.

- ❏ DCL is the gatekeeper of your database.



TCL — Transaction Control Language

Purpose

TCL commands manage **transactions in the database** — ensuring that a group of operations either all succeed or all fail together, maintaining data integrity.

- COMMIT — permanently saves all changes in a transaction
- ROLLBACK — undoes all changes since last commit
- SAVEPOINT — sets a recovery point within a transaction

Real-World Example

During exam result entry: If marks are updated for 50 students and an error occurs at student 30, a *ROLLBACK* reverses all changes. Once verified, a *COMMIT* permanently saves the data.



TCL is your safety net — undo what went wrong.

DDL vs DML

Aspect	DDL	DML
Full Form	Data Definition Language	Data Manipulation Language
Operates On	Structure / Schema of the database	Data / Rows inside tables
Key Commands	CREATE, ALTER, DROP, TRUNCATE, RENAME	INSERT, UPDATE, DELETE
Reversible?	No — auto-committed immediately	Yes — can be rolled back via TCL
Analogy	Building the shelves in a library	Placing and rearranging books on those shelves

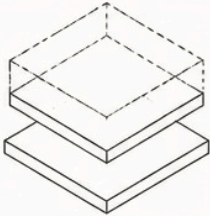
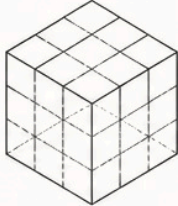
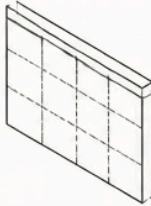
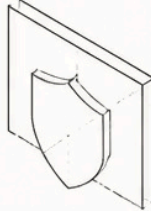
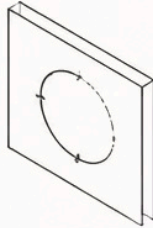
DQL vs DML

Aspect	DQL	DML
Full Form	Data Query Language	Data Manipulation Language
Action	Reads / fetches data only	Writes / modifies data
Key Commands	SELECT	INSERT, UPDATE, DELETE
Changes Data?	No — purely read-only	Yes — alters stored records
Analogy	Reading a book from a shelf	Writing in, tearing pages from, or adding to a book

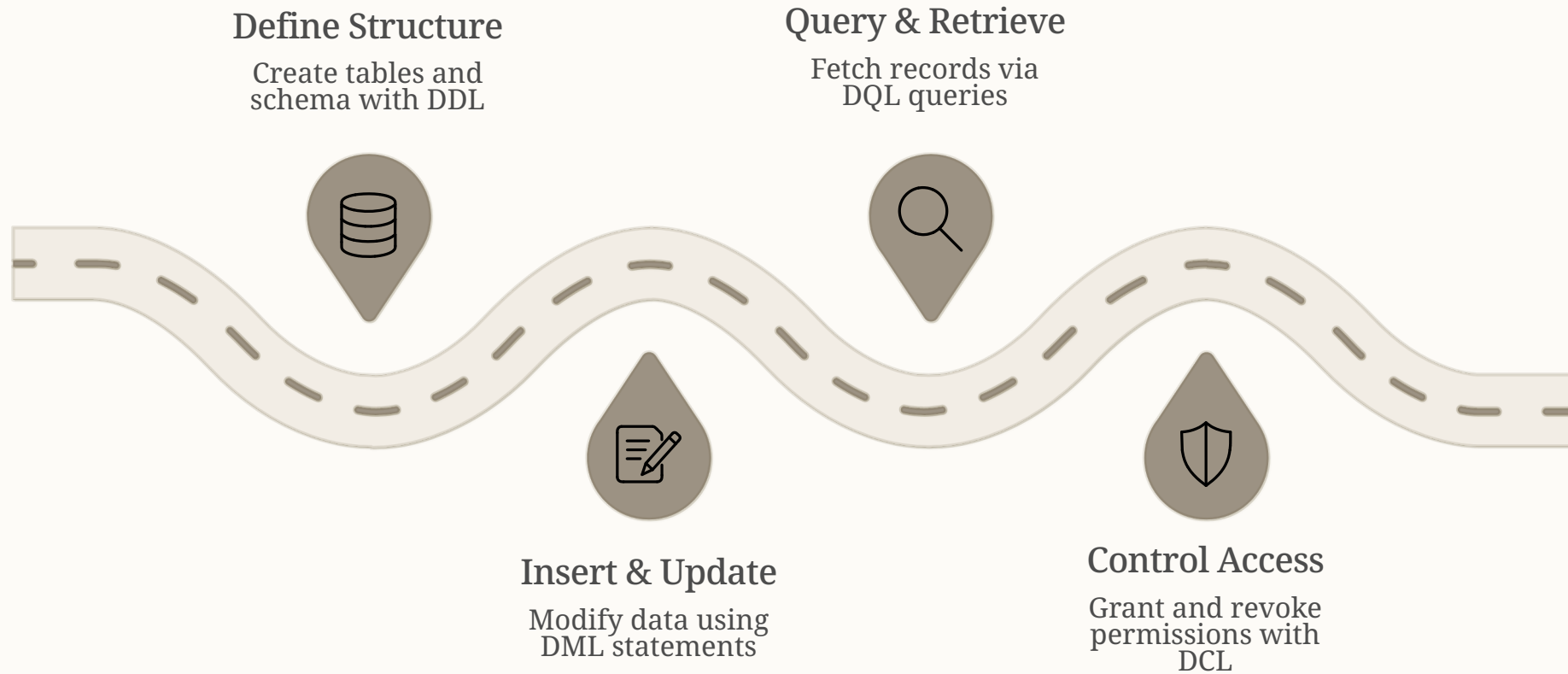
DCL vs TCL

Aspect	DCL	TCL
Full Form	Data Control Language	Transaction Control Language
Purpose	Controls user permissions and access rights	Controls the flow and safety of transactions
Key Commands	GRANT, REVOKE	COMMIT, ROLLBACK, SAVEPOINT
Who Uses It?	Database Administrators (DBAs)	Developers and DBAs managing transactions
Analogy	Giving someone a key to a room	Deciding whether to lock a door after entering

Complete Comparison: All Five Categories

DDL	DML	DQL	DCL	TCL
				
CREATE, ALTER, DROP, TRUNCATE, RENAME	INSERT, UPDATE, DELETE	SELECT	GRANT, REVOKE	COMMIT, ROLLBACK, SAVEPOINT
defines structure - not reversible	modifies data - reversible	retrieves data - read only	controls access - by DBA	manages transactions - safety net

SQL Command Workflow



Every database operation follows this natural progression — from building the structure all the way to securing and finalizing your data operations.

Case Study: Student Management System



DDL — Build the System

Create tables for Students, Courses, and Marks. Add columns as needs grow.



DML — Populate & Update

Enroll students, update contact info, delete withdrawn students.



DQL — Generate Reports

Fetch student marks, course rosters, and attendance summaries.



DCL — Set Permissions

Grant faculty write access; give students read-only view of their own data.



TCL — Commit Results

After entering exam marks for all students, COMMIT to finalize, or ROLLBACK on error.

Common Mistakes Beginners Make

Confusing DELETE with DROP

DELETE (DML) removes rows and can be rolled back.
DROP (DDL) destroys the entire table permanently.

Treating SELECT as DML

SELECT only reads data — it belongs to DQL, not DML.
It never modifies the database.

Forgetting that DDL auto-commits

DDL changes are immediate and irreversible. You cannot ROLLBACK a DROP or TRUNCATE.

Mixing up TRUNCATE and DELETE

TRUNCATE is DDL (removes all rows, no rollback).
DELETE is DML (can be selective and reversed).

Classroom Activity

Identify the correct SQL command category for each task below. Discuss in pairs!

1

Add a new "Phone" column to the Students table

Which category? DDL / DML / DQL / DCL / TCL

2

Retrieve all students who failed an exam

Which category? DDL / DML / DQL / DCL / TCL

3

Allow a librarian to view the Books table only

Which category? DDL / DML / DQL / DCL / TCL

4

Undo a batch of mark entries after an error

Which category? DDL / DML / DQL / DCL / TCL



Quick Quiz + Assignment

5-Question MCQ Quiz

- 1 Which command belongs to DDL? (a) INSERT (b) SELECT (c) CREATE (d) COMMIT
- 2 Which category is used to retrieve data? (a) DDL (b) DML (c) DQL (d) DCL
- 3 ROLLBACK belongs to which category? (a) DML (b) TCL (c) DCL (d) DDL
- 4 GRANT and REVOKE are commands of: (a) DQL (b) TCL (c) DCL (d) DML
- 5 Which DDL command removes all rows but keeps the table? (a) DROP (b) DELETE (c) TRUNCATE (d) ALTER

Assignment

Think of a **Library Management System**. Write down **one real-world task** for each SQL category (DDL, DML, DQL, DCL, TCL) that would be performed in that system.

Submit as a handwritten note or typed document in the next class.



Answers: C, C, B, C, C

Key Takeaways

5 Categories

DDL, DML, DQL, DCL, TCL — each with a distinct purpose

Structure vs Data

DDL defines the skeleton; DML and DQL manage and read the content

Safety Rules

DDL is irreversible. DML can be rolled back. Know the difference!

Access + Integrity

DCL controls who enters; TCL ensures transactions are complete and correct

NEXT LECTURE

Lecture 4: DDL Commands in Depth

We will explore **CREATE**, **ALTER**, **DROP**, **TRUNCATE**, and **RENAME** with syntax, examples, and hands-on practice in MySQL Workbench.